

Total No. of Questions : 8]

SEAT No. :

P3226

[5461]-267

[Total No. of Pages : 2

B.E. (Computer)

DATA MINING TECHNIQUES AND APPLICATIONS
(2012 Pattern) (Semester - I) (End Sem.) (Elective - I) (410444D)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) Use the two methods below to normalize the following group of data :

200, 300, 400, 600, 1000

[6]

i) min-max normalization by setting min = 0 and max = 1

ii) z-score normalization

b) A database has five transactions. Let min sup = 60% and min conf = 80%.

[6]

TID	Items
T100	{M, O, N, K, E, Y}
T200	{D, O, N, K, E, Y}
T300	{M, A, K, E}
T400	{M, U, C, K, Y}
T500	{C, O, O, K, I, E}

Calculate all frequent item sets using Apriori algorithm.

c) Write short note on :

[8]

i) Extracting Rules from decision trees.

ii) KNN approach

OR

Q2) a) Explain data preprocessing steps in short.

[8]

b) Explain performance metrics : Accuracy, Precision, Recall and F-Measure with required equations.

[8]

c) What do you mean by frequent patterns? Explain constraint based association rule based mining frequent item sets.

[4]

P.T.O.

- Q3)** a) Write equations defining Manhattan, Minkowski and Euclidean distance measures. [6]
b) Explain AGNES and DIANA (Agglomerative and Divisive Hierarchical Clustering). [5]
c) Write and explain K-means clustering algorithm. [6]

OR

- Q4)** a) Explain k-Medoid Clustering algorithm. [8]
b) Explain how k-Medoid is extended to CLARANS for handling large data sets. [4]
c) What are typical requirements of clustering in data mining (explain any five)? [5]

- Q5)** a) Explain basic measures for text retrieval like Precision, Recall etc. in the terms of retrieved and relevant documents. [8]
b) Compare: Web content mining and Web usage mining. [6]
c) What are methods for Dimensionality Reduction of Text in text mining? List the methods. [3]

OR

- Q6)** a) Compare different text mining approaches. [5]
b) Write and explain Hyperlink-Induced Topic (HITS) algorithm. [6]
c) Explain these terms in short: Feature vector, Bag of words, Tf-Idf. [6]

- Q7)** a) Write notes on [10]
i) Reinforcement learning,
ii) Systematic Learning
b) Write a note on multi-perspective learning along with diagram for multi-perspective learning. [6]

OR

- Q8)** a) What is Big Data? How it is characterized? What are the challenges in Big data Analysis? [8]
b) What are techniques for big data mining? Explain in short. [8]

